

AGML Lab: Week 7

Bagging and Boosting



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(Jupyter Notebook)

Our model could have high bias or high variance.

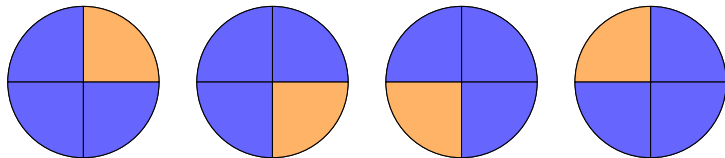
- ▶ High variance $\rightarrow$ Bagging
- ▶ High bias $\rightarrow$ Boosting

But first: a better evaluation of our models.

Motivation: a regular train/test split might result in an unlucky split, that is not representative of the data. $\rightarrow$ get many splits and average the results.

1. Shuffle the data
2. Split the data into k folds
3. For each fold: train on the other $k - 1$ folds, test on this fold
4. Average the results

k -fold cross-validation with $k = 4$, train on blue, test on orange:

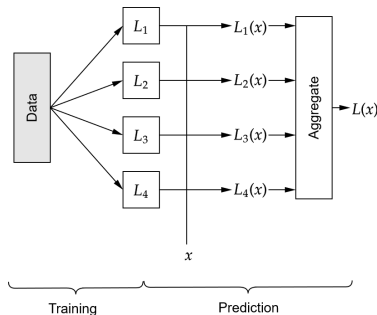


Our model has high variance.

- ▶ Train n models on n different bootstrap dataset (in parallel)
- ▶ For prediction, aggregate the results of the n models (e.g. majority vote).

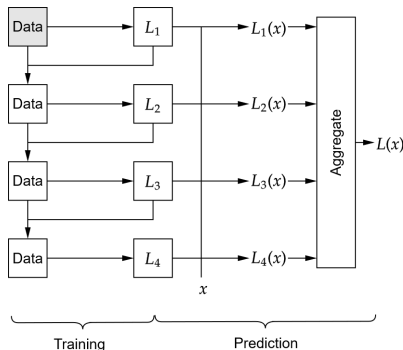
Bootstrap dataset:

- ▶ sample from original dataset with replacement
- ▶ originally same size as original dataset, but this is not necessary (we can also choose to sample less data points)



Our model has high bias.

- ▶ Train n models sequentially
- ▶ Each model tries to correct the mistakes of the previous model
- ▶ For prediction, aggregate the results of the n models



In order to fix the mistakes of the previous model, we introduce weights to the data points.

- ▶ initially, all data points have the same weight
- ▶ after each model, we increase the weight of the misclassified data points
- ▶ the next model will focus more on these data points

A measure of how good a model performs is then a weighted accuracy:

$$\text{Acc}_w = \frac{\sum_{i=1}^m w_i \cdot \mathbb{1}[L(x_i) = y_i]}{\sum_{i=1}^m w_i}$$

Then we update the weights with:

$$w_i \leftarrow w_i \cdot \exp \left(\underbrace{\log \left(\frac{\text{Acc}_w}{1 - \text{Acc}_w} \right)}_{\alpha} \cdot \mathbb{1}[L(x_i) \neq y_i] \right)$$

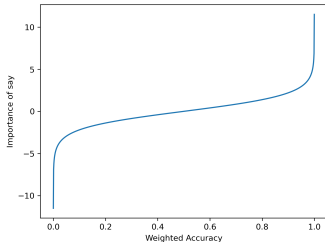
where α is also called the importance of say.

$$\alpha = \log \left(\frac{\text{Acc}_w}{1 - \text{Acc}_w} \right) \quad w_i \leftarrow w_i \cdot \exp(\alpha \cdot \mathbb{1}[L(x_i) \neq y_i])$$

- If we have a very good classifier, we have a large α and we increase the weight of misclassified samples drastically.
- If we have a very bad classifier with large negative α , we actually have a good classifier that just flips the labels wrong. Hence we decrease the weight of these samples.

Use α to aggregate the prediction of the k models:

$$\hat{y}_i = \mathbb{1} \left[\left(\sum_k \alpha_k \cdot (2L_k(x_i) - 1) \right) > 0 \right]$$



(Jupyter Notebook)

- ▶ How could we aggregate the predictions for regression instead of classification?
- ▶ How would you link Bagging and Boosting to kebabs?

Course Evaluation

Course:

„Algorithmische Grundlagen des Maschinellen Lernens LAB
(Statistische Lerntheorie)“

Lecturer:

Prof. Dr. Joachim Giesen et al.

This QR code will take you to the evaluation for the course mentioned above.

Alternatively, please open the following website:

www.evaluation.uni-jena.de/lve/

and log in there with the following data:

Survey-ID: 40571

Password: 39100b

